Ex. No. 5	**Subqueries and Correlated Subqueries**	
Date of Exercise		

Objective:

To execute given query using the concept of subqueries and correlated subqueries.

Description:

Subquery is usually added in the WHERE Clause of the sql statement. Most of the time, a subquery is used when you know how to search for a value using a SELECT statement, but do not know the exact value. Subqueries are an alternate way of returning data from multiple tables. Subqueries can be used with the following sql statements along with the comparison operators like =, <, >, >=, <= etc.

- Update
- Insert
- Select
- Delete

Correlated sub-query

A **correlated sub-query** is a term used for specific types of queries in SQL in computer databases. It is a sub-query (a query nested inside another query) that uses values from the outer query in its WHERE clause. The sub-query is evaluated once for each row processed by the outer query.

Detailed Procedure:

Here is an example for a typical correlated sub-query. In this example we are finding the list of employees (employee number and names) having more salary than the average salary of all employees in that employee's department.

```
SELECT employee_number, name FROM employee AS e1
WHERE salary > (SELECT avg(salary)FROM employee
WHERE department = e1.department);
```

In the above query the outer query is,

```
SELECT employee_number, name FROM employee AS e1
 WHERE salary >
```

And the inner query is,

```
(SELECT avg(salary)
FROM employee
WHERE department = e1.department);
```

In the above nested query, the inner query has to be executed for every employee as the department will change for every row. Hence the average salary will also change. The effect of correlated sub-queries can also be obtained using outer Joins.

Sample Input and Output:

Tables: -

Create the following tables

Create table customer (cust_id number (3), cust_name varchar2(10), annual_revenue number (10), cust_type varchar2(12));
Commit;
Create table SHIPMENT (SHIPMENT_# number(3), CUST_ID number(3) , WEIGHT number(4,1), TRUCK_# number(3), DESTINATION varchar2(15),
SHIP_DATE date);
Create table TRUCK (TRUCK_# number(3), DRIVER_NAME varchar2(15));
Create table CITY (CITY_NAME varchar2(10), POPULATION number(15));
Solve the following complex SQL Queries using Exists or NOT Exists

1. What are the names of customers who have sent packages (shipments) to Sioux City?
   CUST_NAME

   Revathi
2. To what destinations have companies with revenue less than $1 million sent packages?
   DESTINATION

   London
   Los Angeles
   Sioux City
3. What are the names and populations of cities that have received shipments weighing over 100 pounds?

CITY_NAME	POPULATION
London	100000000
Manhattan	10000000
Rome	200000000
Sioux City	5000000000

4.Who are the customers having over $5 million in annual revenue who have sent shipments weighing less than 1 pound?

CUST_ID
311

5.Who are the customers having over $5 million in annual revenue who have sent shipments weighing less than 1 pound or have sent a shipment to San Francisco?
No row Selected

QUESTIONS AND QUERIES:

1. Retrieve users who have reserved for a sports tournament.

2. Find events hosted in London.
3. Calculate average ticket price for an event Art Exhibition.
4. Retrieve users who haven't made any reservations.
5. Calculate the maximum price among all events reserved by a user.
6. Find users who have reserved tickets for events on 20th July 2023.
7. Find users who have reserved tickets for events at USA.
8. Retrieve events where the price of the events is less than $.30.
9. Calculate the total revenue generated by Ethan.
10. Find the events which have been conducted at the same venue.

RESULT: